Pairwise Character Geometry and the Classification of Neutral Generators in SU(2)

Step 3 of the Admissibility Sub-Programme

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Abstract

We study the geometry of *neutral generating sets*: symmetric generating sets $S = S^{-1}$ of a finite group G equipped with an irreducible representation $\rho\colon G \to \mathrm{SU}(2)$ satisfying $\chi_\rho(s) = 0$ for all $s \in S$. The central result is that the geometry of the associated unit axes $\{\vec{u}_s\}_{s\in S} \subset S^2$ is entirely encoded in the pairwise character values: $G_{st} := \vec{u}_s \cdot \vec{u}_t = -\frac{1}{2}\chi_\rho(st)$. Using this identity together with the second-moment tensor $M = \sum_{s\in S^+} \vec{u}_s \vec{u}_s^T$, we give a complete character-theoretic classification of all admissible groups:

(i) $M \not\propto I \implies \rho(G) \cong \mathrm{Dic}_n$, $n \geq 3$;

(ii) $M \propto I$, $G_{st} = 0$ for all s, t with $t \neq s^{\pm 1} \implies \rho(G) \cong Q_8$;

(iii) $M \propto I$, $G_{st} \notin \{0, \pm 1\}$ for some $s, t \implies \rho(G) \in \{2O, 2I\}$.

In particular, the binary tetrahedral group $2T$ (projective image A_4) admits no neutral generating set, a structural exclusion we prove directly. The finite subgroups of SU(2) compatible with spectral neutrality are exactly Q_8, Dic_n ($n \geq 3$), $2O$, and $2I$. This completes the Admissibility Sub-Programme of Cosmochrony.

Keywords: spectral admissibility, Cayley graphs, Gram matrices, representation theory, finite subgroups of SU(2), binary polyhedral groups, quaternion group Q8

Contents

1 Introduction

The Admissibility Sub-Programme of Cosmochrony asks which spectra of the relational operator $\mathcal{L}_\chi$ are compatible with the bounded-flux evolution constraint $\|c_\chi\| \le 1$. The programme proceeds in three steps.

- **Step 1** (*SpectralAdmissibility 1.0*): generators satisfying the neutrality condition $\chi_\rho(s) = 0$ correspond to traceless elements of SU(2), i.e. to half-turns.
- **Step 2** (*SpectralCapacity 1.0*): the Klein classification of finite subgroups of SU(2) restricts the possible groups $\rho(G)$ to $\{\mathrm{Dic}_n\ (n \ge 2),\ 2T,\ 2O,\ 2I\}$.
- **Step 3** (*this paper*): pairwise character geometry distinguishes the cases analytically and establishes which groups actually arise.

The key difficulty left open after Step 2 is that the Klein classification is typically proved by geometric or group-theoretic means, neither of which refers to the spectral constraint. The main contribution of the present paper is to show that the full classification *follows* from two character-theoretic invariants: the second-moment tensor M and the Gram matrix G_{st}. No geometric input beyond neutrality is needed.

Main result (informal). The finite subgroups of SU(2) admissible under spectral neutrality are exactly

$$Q_8, \quad \mathrm{Dic}_n\ (n \ge 3), \quad 2O, \quad 2I.$$

The binary tetrahedral group $2T$ is *structurally excluded*: the full set of its traceless elements generates only Q_8, not $2T$. The three remaining cases are separated by whether $M \propto I$ (isotropy) and by whether the Gram matrix contains values outside $\{0, \pm 1\}$.

Organisation. Section 2 recalls the setup and the results of Steps 1–2. Section 3 establishes the Gram identity $G_{st} = -\frac{1}{2}\chi_\rho(st)$. Section 4 analyses the second-moment tensor and proves that Dic_n is anisotropic for $n \ge 3$. Section 5 proves that an orthogonal Gram pattern forces $\rho(G) \cong Q_8$. Section 6 proves the structural exclusion of $2T$. Section 7 assembles the main classification theorem. Section 8 discusses completeness via the Klein list. Section 9 draws the conceptual conclusions.

2 Setup: neutral generators in SU(2)

Let G be a finite group, $S = S^{-1} \subset G$ a symmetric generating set, and $\rho\colon G \to \mathrm{SU}(2)$ an irreducible two-dimensional representation. The *neutrality condition* is

$$\chi_\rho(s) \;:=\; \mathrm{Tr}\,\rho(s) \;=\; 0 \qquad \forall\, s \in S. \tag{1}$$

Write $S^+ \subset S$ for a fixed choice of one representative from each pair $\{s, s^{-1}\}$.

Step-1 recalled. Since $\rho(s) \in \mathrm{SU}(2)$ is traceless, its characteristic polynomial is $\lambda^2 + 1 = 0$, so $\rho(s)^2 = -I$. Every traceless element of SU(2) can be written

$$\rho(s) \;=\; i\,(\vec{u}_s \cdot \vec{\sigma}), \qquad \vec{u}_s \in S^2 \subset \mathbb{R}^3, \tag{2}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices and the sign is fixed by $\rho(s)^2 = -I$. (Replacing $\rho(s)$ by $-\rho(s)$ replaces $\vec{u}_s$ by $-\vec{u}_s$; both choices yield the same half-turn in SO(3).) Thus each $s \in S$ determines a unique axis $\pm\vec{u}_s \in \mathbb{RP}^2$.

Step-2 recalled. By (2), every $\rho(s)$ has order 4 in SU(2), so its image $\bar{\rho}(s)$ in SO(3) is a half-turn. The Klein classification of finite subgroups of SU(2) (see, e.g., [1]) gives the complete list

$$\mathbb{Z}_n, \quad \mathrm{Dic}_n\ (n \geq 1), \quad 2T, \quad 2O, \quad 2I,$$

with projective images $\mathbb{Z}_n$, D_n, A_4, S_4, A_5 in SO(3) respectively. Since the cyclic groups $\mathbb{Z}_n$ contain no element of order 4 for $n \leq 2$, and no traceless element at all for $n \geq 3$, the neutrality condition (1) reduces the candidates to

$$\rho(G) \ \in \ \big\{\mathrm{Dic}_n\ (n \geq 2),\ 2T,\ 2O,\ 2I\big\}, \qquad \mathrm{Dic}_2 \cong Q_8. \tag{3}$$

3 Pairwise character geometry: the Gram identity

Lemma 3.1 (Gram identity)**.** *For any $s, t \in S$, let $G_{st} := \vec{u}_s \cdot \vec{u}_t$ be the Euclidean dot product of the corresponding unit axes in $\mathbb{R}^3$. Then*

$$G_{st} \ = \ -\tfrac{1}{2}\,\chi_\rho(st). \tag{4}$$

Proof. Using the Pauli product identity $(\vec{u}\cdot\vec{\sigma})(\vec{v}\cdot\vec{\sigma}) = (\vec{u}\cdot\vec{v})\,I + i\,(\vec{u}\times\vec{v})\cdot\vec{\sigma}$, we compute

$$\begin{aligned}\rho(s)\,\rho(t) = \big(i\,\vec{u}_s\cdot\vec{\sigma}\big)\big(i\,\vec{u}_t\cdot\vec{\sigma}\big) \ &= \ -\big(\vec{u}_s\cdot\vec{\sigma}\big)\big(\vec{u}_t\cdot\vec{\sigma}\big)\\ &= -(\vec{u}_s\cdot\vec{u}_t)\,I \ - \ i\,(\vec{u}_s\times\vec{u}_t)\cdot\vec{\sigma}.\end{aligned}$$

Taking the trace and using $\mathrm{Tr}(\vec{w}\cdot\vec{\sigma}) = 0$:

$$\chi_\rho(st) \ = \ \mathrm{Tr}\big(\rho(s)\rho(t)\big) \ = \ -2\,G_{st}. \qquad \square$$

Remark 3.2. Lemma 3.1 is the conceptual core of the paper. It shows that the *entire axis geometry* — the pairwise angles between neutral generators in S^2 — is readable directly from the character table: G_{st} is computable from χ_ρ alone, without any geometric input. In particular, the Gram matrix $(G_{st})_{s,t\in S}$ is an intrinsic spectral invariant of the representation ρ.

4 Second-moment tensor and the dihedral case

Definition 4.1. *The* second-moment tensor *of the generator set S is*

$$M \ := \ \sum_{s\in S^+} \vec{u}_s\,\vec{u}_s^{\,T} \ \in \ \mathrm{Sym}_3(\mathbb{R}). \tag{5}$$

By construction M is positive semidefinite. Its isotropy class — whether $M \propto I$ or not — is the first character-theoretic discriminant in the classification.

Proposition 4.2 (Dihedral anisotropy)**.** *Let $\rho(G) \cong \mathrm{Dic}_n$ with $n \geq 3$, and let S be the full set of traceless elements of G. In the basis where $\vec{e}_z$ is the C_{2n} symmetry axis, the second-moment tensor M has eigenvalues*

$$\mathrm{spec}(M) \ = \ \begin{cases}(n,\ n,\ 0) & \text{if } n \text{ is odd,}\\ (n,\ n+2,\ 0) & \text{if } n \text{ is even.}\end{cases} \tag{6}$$

In particular, $M \not\propto I$ for all $n \geq 3$.

Proof. In the standard presentation, Dic_n has a cyclic normal subgroup $\langle a\rangle \cong \mathbb{Z}_{2n}$ generated by $a = (\cos(\pi/n),\, \sin(\pi/n),\, 0,\, 0)$ and an order-4 element $b = (0,\, 0,\, 1,\, 0)$ satisfying $bab^{-1} = a^{-1}$. We identify quaternions with (a_0, a_1, a_2, a_3) and the corresponding axis via $(0, a_1, a_2, a_3) \mapsto \vec{u} = (a_2, a_3, a_1)$.

(A) Flip-coset elements. A direct computation gives

$$b\,a^k \;=\; (0,\;\sin(k\pi/n),\;\cos(k\pi/n),\;0), \qquad k = 0,\ldots,2n-1, \tag{7}$$

hence unit axes $\vec{u}_k = (\cos(k\pi/n),\,0,\,\sin(k\pi/n))$. All $2n$ flip-coset axes are uniformly distributed in the xz-plane. Using the standard identities for $2n$ uniform points on a circle,

$$\sum_{k=0}^{2n-1} \cos^2 \tfrac{k\pi}{n} = n, \quad \sum_{k=0}^{2n-1} \sin^2 \tfrac{k\pi}{n} = n, \quad \sum_{k=0}^{2n-1} \cos \tfrac{k\pi}{n} \sin \tfrac{k\pi}{n} = 0,$$

the flip-coset contribution to M is

$$M_{\text{flip}} \;=\; \sum_{k=0}^{2n-1} \vec{u}_k\,\vec{u}_k^T \;=\; \operatorname{diag}(n,\;0,\;n).$$

(B) Cyclic traceless elements (n even only). The cyclic element a^k has real part $\cos(k\pi/n)$, which vanishes exactly for $k = n/2$ and $k = 3n/2$ (when n is even, and not otherwise). These two elements are $a^{n/2} = (0,1,0,0) = \pm i$ with axis $\vec{e}_z = (0,0,1)$, contributing $2\,\vec{e}_z\vec{e}_z^T = \operatorname{diag}(0,0,2)$.

(C) Total. Combining both contributions:

$$M \;=\; \begin{cases} \operatorname{diag}(n,\;0,\;n) & n \text{ odd (no cyclic traceless elements)}, \\ \operatorname{diag}(n,\;0,\;n+2) & n \text{ even (two cyclic traceless elements } \pm i). \end{cases}$$

In both cases the eigenvalues are $(n,\,0,\,n)$ or $(n,\,0,\,n+2)$, i.e. the spectrum stated in (6) (reordering the diagonal). Since the zero eigenvalue is strictly less than $n \geq 3$, we have $M \not\propto I$. □

Remark 4.3. The special case $n = 2$ (i.e. $\mathrm{Dic}_2 \cong Q_8$) is *isotropic*: the six traceless elements $\{\pm i, \pm j, \pm k\}$ span all three coordinate axes, giving $M_{Q_8} = 2I$. This is not contradicted by (6): the formula applies to Dic_n for $n \geq 2$, but for $Q_8 = \mathrm{Dic}_2$ the cyclic subgroup contributes traceless elements $\pm i$ (axis $\vec{e}_z$) *and* the flip coset contributes $\pm j, \pm k$ (axes $\vec{e}_x, \vec{e}_y$), restoring full isotropy. Proposition 4.2 is thus sharp: the anisotropy holds exactly for $n \geq 3$.

5 The orthogonal Gram lemma: Q_8

Lemma 5.1 (Orthogonal Gram $\Rightarrow Q_8$). *Suppose $S = S^{-1}$ generates G, the neutrality condition* (1) *holds, ρ is irreducible, and*

$$G_{st} = 0 \qquad \forall\, s,t \in S \text{ with } t \neq s^{\pm 1}. \tag{8}$$

Then $\rho(G) \cong Q_8$.

Proof. **Step A (at most three distinct axis directions).** Condition (8) says any two generators not in the same inverse pair have $\vec{u}_s \perp \vec{u}_t$ in $\mathbb{R}^3$. A set of mutually orthogonal unit vectors in $\mathbb{R}^3$ has at most three elements. Denote the distinct axis directions $\vec{e}_1, \vec{e}_2, \vec{e}_3 \in S^2$ (at most three, not all need be present).

Step B (at least two distinct directions). If all generators share the same axis $\pm\vec{e}_1$, then $\rho(G) \subset \{\pm I, \pm i\,\vec{e}_1 \cdot \vec{\sigma}\} \cong \mathbb{Z}_4$, which is abelian, contradicting irreducibility of ρ. Hence there are at least two distinct axis directions.

Step C (cross product generates the third direction). Let $\vec{e}_1, \vec{e}_2$ be two distinct orthogonal axis directions. The product $\rho(s_1)\rho(s_2)$, where $\vec{u}_{s_1} = \vec{e}_1$ and $\vec{u}_{s_2} = \vec{e}_2$, is

$$\begin{aligned}\rho(s_1)\rho(s_2) = (i\,\vec{e}_1 \cdot \vec{\sigma})(i\,\vec{e}_2 \cdot \vec{\sigma}) \;&=\; -(\vec{e}_1 \cdot \vec{e}_2)\, I - i(\vec{e}_1 \times \vec{e}_2) \cdot \vec{\sigma} \\ &= -i\,(\vec{e}_1 \times \vec{e}_2) \cdot \vec{\sigma},\end{aligned}$$

since $\vec{e}_1 \perp \vec{e}_2$. This is traceless with axis $\vec{e}_3 := \vec{e}_1 \times \vec{e}_2$. Hence the product $\rho(s_1 s_2)$ introduces the axis $\vec{e}_3$, and the three directions $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ form a right-handed orthonormal frame.

Step D (closure and identification). The set of eight elements

$$\{\pm I,\ \pm i\,\vec{e}_1 \cdot \vec{\sigma},\ \pm i\,\vec{e}_2 \cdot \vec{\sigma},\ \pm i\,\vec{e}_3 \cdot \vec{\sigma}\}$$

is closed under multiplication by the Pauli product identity (using $\vec{e}_a \times \vec{e}_b = \pm\vec{e}_c$ for the cyclic triples), hence forms a subgroup of SU(2) isomorphic to Q_8. Since all generators of S belong to this set and S generates G, we have $\rho(G) \subset Q_8$; irreducibility forces $\rho(G) = Q_8$. $\square$

Remark 5.2. The condition $t \neq s^{\pm 1}$ in (8) is necessary. For any $s \in S$, we have $G_{s,s^{-1}} = \vec{u}_s \cdot (-\vec{u}_s) = -1$, which belongs to $\{0, \pm 1\}$ but is not zero. The lemma asserts that all *off-diagonal* (non-inverse-pair) entries vanish.

6 Structural exclusion of $2T$

Proposition 6.1 (Exclusion of $2T$)**.** *The binary tetrahedral group $2T$ admits no symmetric neutral generating set. Consequently, A_4 cannot appear as the projective image $\bar{\rho}(G)$ of any neutral generating set in an irreducible 2-dimensional representation.*

Proof. The group $2T$ has order 24 and decomposes as

$$\text{Coset 1 } (Q_8):\quad \{\pm 1,\ \pm i,\ \pm j,\ \pm k\}, \tag{9}$$

$$\text{Coset 2:}\quad \left\{\tfrac{\pm 1 \pm i \pm j \pm k}{2}\right\} \quad \text{(all 16 sign combinations).} \tag{10}$$

In the spin-$\frac{1}{2}$ representation, the character of a quaternion $q = a + bi + cj + dk$ is $\chi(q) = \mathrm{Tr}(\rho(q)) = 2a$. Hence $\chi(q) = 0$ if and only if $a = 0$. Among elements of Coset 1, $a = 0$ only for $\{\pm i, \pm j, \pm k\}$ (since ± 1 have $a = \pm 1$). Every element of Coset 2 has $|a| = \frac{1}{2} \neq 0$.

Therefore the set of all traceless elements of $2T$ is exactly $T_0 := \{\pm i, \pm j, \pm k\} = Q_8 \setminus \{\pm 1\}$. This set generates Q_8 (order 8), not $2T$ (order 24). Any symmetric neutral generating set for $2T$ must consist of traceless elements, hence $\langle S \rangle \subset \langle T_0 \rangle = Q_8 \subsetneq 2T$, so S cannot generate $2T$. $\square$

Remark 6.2. The exclusion of $2T$ is a structural consequence of the quaternionic coset decomposition: the non-identity elements of the cyclic part $\langle a \rangle$ in Dic_n (for $n \geq 2$) are traceless only at specific orders, but in $2T$ the entire second coset has $|a| = 1/2$, placing it irreversibly outside the neutral sector.

7 Main theorem: Gram rigidity

Theorem 7.1 (Gram rigidity of neutral generators)**.** *Let G be a finite group, $S = S^{-1} \subset G$ a symmetric generating set, and $\rho\colon G \to \mathrm{SU}(2)$ an irreducible two-dimensional representation satisfying $\chi_\rho(s) = 0$ for all $s \in S$. Define*

$$M \;:=\; \sum_{s \in S^+} \vec{u}_s\, \vec{u}_s^{T}, \qquad G_{st} \;:=\; -\tfrac{1}{2}\,\chi_\rho(st) \;=\; \vec{u}_s \cdot \vec{u}_t.$$

Then exactly one of the following cases holds.

(i) $M \not\propto I$*: then* $\bar{\rho}(G) \cong D_n$ *for some* $n \geq 3$*, i.e.* $\rho(G) \cong \mathrm{Dic}_n$*.*

(ii) $M \propto I$ *and* $G_{st} = 0$ *for all* $s, t \in S$ *with* $t \neq s^{\pm 1}$*: then* $\rho(G) \cong Q_8$*.*

(iii) $M \propto I$ *and* $G_{st} \notin \{0, \pm 1\}$ *for some* $s, t \in S$*: then* $\rho(G) \in \{2O, 2I\}$*.*

Moreover, the binary tetrahedral group $2T$ *(projective image* A_4*) does not appear in any of the three cases.*

Proof. **Mutual exclusivity.** Cases (i)–(iii) are mutually exclusive by definition: $M \not\propto I$ versus $M \propto I$, then the Gram values.

Exhaustiveness. Every finite group G satisfying the hypotheses has $\rho(G)$ in the list $\{\mathrm{Dic}_n, 2T, 2O, 2I\}$ by Step 2 (3). Proposition 6.1 excludes $2T$. Proposition 4.2 shows Dic_n ($n \geq 3$) satisfies $M \not\propto I$, hence falls in case (i). $Q_8 = \mathrm{Dic}_2$ has $M = 2I$ (the six traceless elements $\{\pm i, \pm j, \pm k\}$ span all three coordinate axes; see Remark 4.3) and all off-diagonal Gram values in $\{0, -1\}$ (since $\{\pm i, \pm j, \pm k\}$ are pairwise orthogonal), so it falls in case (ii). The remaining candidates $2O$ and $2I$ have $M \propto I$ (by the isotropy remark below) and Gram values outside $\{0, \pm 1\}$, so they fall in case (iii).

Case (i) conclusion. By Proposition 4.2, Dic_n with $n \geq 3$ is the unique possibility, and $\bar{\rho}(\mathrm{Dic}_n) = D_n$.

Case (ii) conclusion. With $M \propto I$ and all off-diagonal Gram entries zero, the hypotheses of Lemma 5.1 are met, giving $\rho(G) \cong Q_8$.

Case (iii) conclusion. With $M \propto I$ and a non-trivial Gram value, $\rho(G)$ is neither Dic_n (excluded by isotropy and Proposition 4.2), nor Q_8 (excluded by the non-trivial Gram value), nor $2T$ (excluded by Proposition 6.1). By (3), the only remaining possibilities are $2O$ and $2I$. Both cases occur: taking S to be the full set of traceless elements of $2O$ (resp. $2I$), which does form a symmetric neutral generating set for those groups, one finds $M \propto I$ (see Remark 7.2) and Gram values $\cos(\pi/4), \cos(\pi/3)$ for $2O$ and $\cos(\pi/5), \cos(\pi/3), \cos(2\pi/5)$ for $2I$. ☐

Remark 7.2 (Isotropy of the full traceless sets for $2T$, $2O$, $2I$). Let $H \in \{A_4, S_4, A_5\}$ act on $\mathbb{R}^3$ through its standard rotational representation, and let $\mathcal{U} \subset S^2$ be any H-stable finite set of unit axes. Define $M_{\mathcal{U}} := \sum_{\vec{u} \in \mathcal{U}} \vec{u}\,\vec{u}^T$. For any $h \in H$,

$$h\, M_{\mathcal{U}}\, h^{-1} \;=\; \sum_{\vec{u} \in \mathcal{U}} (h\vec{u})(h\vec{u})^T \;=\; \sum_{\vec{u} \in \mathcal{U}} \vec{u}\,\vec{u}^T \;=\; M_{\mathcal{U}},$$

so $M_{\mathcal{U}}$ lies in the commutant of the H-representation on $\mathbb{R}^3$. Since the standard rotational representations of A_4, S_4, and A_5 on $\mathbb{R}^3$ are irreducible, Schur's lemma forces $M_{\mathcal{U}} = c\, I$ for some scalar $c \geq 0$.

Irreducibility of the A_4*-action.* If a line $L \subset \mathbb{R}^3$ were A_4-invariant, it would be a common fixed axis for all rotational symmetries of the tetrahedron; but the 3-cycles of A_4 correspond to rotations around four distinct axes sharing no common line. Since the representation is orthogonal, an A_4-invariant plane would yield an invariant line as its orthogonal complement; hence no invariant plane exists either. The same argument applies to S_4 (cube/octahedron) and A_5 (icosahedron/dodecahedron).

In particular, the full traceless element sets of $2T$, $2O$, and $2I$ all have isotropic second-moment tensor $M \propto I$, regardless of whether those sets generate the full group or a proper subgroup. For $2T$ specifically, the traceless elements generate only $Q_8 \subset 2T$; isotropy $M \propto I$ still holds, but the generation hypothesis of Theorem 7.1 is not satisfied. These are independent properties.

Remark 7.3 (Separating $2O$ from $2I$ within case (iii)). Theorem 7.1 does not separate $2O$ from $2I$ within case (iii). The distinction is encoded in the actual values of G_{st}: for $2O$, the non-trivial values are $\cos(\pi/4) = 1/\sqrt{2}$ and $\cos(\pi/3) = 1/2$ (octahedral angles); for $2I$, they are

$\cos(\pi/5) = \phi/2$, $\cos(\pi/3) = 1/2$, and $\cos(2\pi/5) = 1/(2\phi)$ (icosahedral angles, $\phi = \frac{1+\sqrt{5}}{2}$). This finer distinction is available from the character table but is not needed for the main admissibility conclusion.

8 Completeness via Klein

The three cases in Theorem 7.1 together cover all finite subgroups of SU(2) admissible under spectral neutrality. The argument is a triple exclusion against the full Klein list:

1. $\mathbb{Z}_n$ is excluded by Step 2: cyclic groups have no traceless elements for $n \geq 3$, and $\mathbb{Z}_1, \mathbb{Z}_2$ cannot be generated by any element.
2. $2T$ is excluded by Proposition 6.1: the full set of traceless elements of $2T$ generates only Q_8, so $2T$ admits no neutral generating set.
3. Dic_n for $n \geq 3$ falls into case (i) and is correctly classified by anisotropy.
4. $Q_8 = \mathrm{Dic}_2$ falls into case (ii).
5. $2O$ and $2I$ fall into case (iii).

No group remains unaccounted for. The admissible finite subgroups of SU(2) under spectral neutrality are exactly

$$Q_8, \quad \mathrm{Dic}_n\ (n \geq 3), \quad 2O, \quad 2I. \tag{11}$$

9 Discussion

The conceptual content of the Gram identity. The proof of Theorem 7.1 rests entirely on two character-theoretic objects: $M = \sum \vec{u}_s \vec{u}_s^T$ and (G_{st}). Both are computable from the character table of G and the representation ρ, with no geometric input. The Gram identity $G_{st} = -\frac{1}{2}\chi_\rho(st)$ is the key step: it shows that the pairwise *angles* between neutral axes are encoded in the *characters of pairwise products*. Geometry emerges from spectral arithmetic.

Character-theoretic reconstruction of the Klein list. The Klein classification of finite subgroups of SU(2) is classically proved by polyhedral geometry or by Dynkin diagram methods. The present paper gives a *character-theoretic* proof of the same classification, restricted to the neutral sector. The three regimes of Theorem 7.1 correspond to:

- an *anisotropy* discriminant ($M \propto I$ or not), detecting the presence of a preferred rotation axis;
- an *orthogonality* discriminant ($G_{st} = 0$ or not), detecting the purely quaternionic geometry;
- a *non-triviality* discriminant ($G_{st} \in \{0, \pm 1\}$ or not), detecting the polyhedral (octahedral or icosahedral) geometry.

Each discriminant is a statement about eigenvalues of products of group elements, hence a purely spectral condition.

Connection to the Admissibility Programme. In the Cosmochrony framework, the generator set S models the flux-carrying modes of the substrate operator $\mathcal{L}_\chi$, and the neutrality condition $\chi_\rho(s) = 0$ encodes compatibility with the bounded-flux constraint $\|c_\chi\| \leq 1$ (Steps 1–2). The present result shows that this constraint is powerful enough to recover the full polyhedral structure of SU(2): the admissible groups (11) are exactly those whose character geometry is consistent with isotropy or orthogonality of the axis configuration.

Structural role of $2T$. The exclusion of $2T$ is not an accident of numerics but a structural consequence of the coset decomposition: the tetrahedral group's neutral sector is entirely contained in Q_8, making it impossible for any neutral generating set to witness the full A_4 symmetry. This asymmetry between $2T$ and the binary octahedral/icosahedral groups $2O$, $2I$ reflects a deeper distinction: for $2O$ and $2I$, the full set of traceless elements *is* a neutral generating set for the group, while for $2T$ it is not.

10 Conclusion

Step 3 of the Admissibility Sub-Programme is now complete. The chain of implications is:

$$\text{neutral generators} \implies \text{pairwise character geometry} \implies \text{finite subgroup structure.}$$

Specifically:

1. The Gram identity (Lemma 3.1) encodes axis geometry in character values.
2. The second-moment tensor (Proposition 4.2) detects the dihedral case by anisotropy.
3. The orthogonal-block lemma (Lemma 5.1) identifies Q_8.
4. The structural exclusion of $2T$ (Proposition 6.1) removes the tetrahedral case.
5. The main theorem (Theorem 7.1) assembles the complete classification (11).

Together with SpectralAdmissibility 1.0 and SpectralCapacity 1.0, this establishes the full picture: the substrate groups compatible with bounded-flux spectral neutrality are exactly Q_8, Dic_n ($n \geq 3$), $2O$, and $2I$.

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